# IQE.L - IQE plc | Full investment artifact

*Full active edition | Financial methods and portfolio Base overrides preserved.*

**Capital update:** 2 October 2026 | **Operating-research date retained:** 17 September 2026 | **Framework:** 8.0 | **Format:** standalone current-state analysis **Status:** NOT HELD / WATCHLIST in supplied original | **Classification:** BOTTLENECK CANDIDATE - qualified compound-semiconductor epitaxy **Artifact Base:** 42/60 = **70% - Watchlist** | **Protected portfolio Base:** 77% (user override retained; separate from artifact assessment) **Snapshot:** supplied-workbook price GBp 47.000; model share proxy 1,412.185691M; implied snapshot market capitalization 663.727 GBP millions. Price is frozen from the supplied workbook, not a current-market refresh; capitalization uses the modeled denominator. **Active ceiling:** Q4 2027 **GBp 47.345615**, pricing Q4 2027-Q3 2028. All 15 quarterly ceilings are derived below.

The opportunity extends across more optical architectures and a larger qualified customer base. The full model allows substantial InP growth while preserving the now-visible operating loss and depreciation burden at the starting point. Customer funding improves execution prospects; it does not guarantee margins or a public-share price floor.

## 1. Supercycle, supply chain and durable advantage

IQE supplies qualified epitaxial materials used in optical and RF devices. Customer agreements and financing strengthen its capacity-conversion opportunity. The central question is whether higher photonics loading can absorb manufacturing depreciation and corporate costs while generating cash. Its strategic review has concluded; current valuation rests on operating execution.

DB1 photonics: optical systems → lasers, modulators and detectors → device fabrication on engineered epi layers → qualified epitaxy grown on raw substrates. IQE is the epi supplier, not the substrate boule grower or finished module vendor. Recipes, uniformity, defect control and customer qualification create switching costs, while competing epitaxy and internal production limit exclusivity. GaAs QDL and InP can serve different or overlapping optical architectures.

| Business / curve | Stage | Evidence and next inflection |
|---|---|---|
| InP optical epitaxy | Early commercial ramp;10-50% qualitative | Signed supply relationships and conversion of tooling support growth. |
| Wireless and legacy photonics | Recovery/mature applications | Handset, infrastructure and defense mix drive utilization. |
| QDL and newer platforms | Pre-ramp 0-10% | Customer sampling must convert into product launches and recurring volume. |

Stage percentages are qualitative framework bands relative to a possible mature opportunity, not measured market penetration. A current capacity constraint can coexist with a larger long-term TAM; qualification and economic capture decide how much reaches this company.

## 2. Forward TAM and the three-phase expansion lens

| Phase | Engineering opportunity | Demand-to-company bridge |
|---|---|---|
| Phase 1;100% thematic | Existing optical/RF epi demand | More optical links and bandwidth create additional wafer demand. IQE captures epi invoice content, not the value of complete transceivers. |
| Phase 2;50% thematic | Higher-speed photonics, OCS and broader customers | Tower's agreement supports a qualified route into next-generation platforms. Modeled InP80M 2027→285M 2030 requires 3.5625×revenue. An illustrative 3×sellable wafer-equivalent volume ×1.1875 content/net ASP gives that endpoint; both factors are analyst assumptions. |
| Phase 3;20% thematic | QDL, microLED, advanced sensing and GaN applications | Physical performance can create new product markets, but qualification, wafer economics and customer demand decide capture. QDL is within other photonics; no full global photonics-TAM overlay is added. |

The 100% / 50% / 20% thematic weights guide thesis assessment. They are not blanket discounts to reported sales, contracts or all forecasts. Programme probabilities below govern only separately identified optionality. Current reported sales calibrate the starting point; they do not cap the future market.

## 3. Current earnings and management evidence

H1 revenue was GBP 64.558M: Photonics 38.535M and Wireless 26.023M. Reported GP/OP were 5.993M/-8.145M; adjusted GP/OP were 6.564M/-3.741M. Adjusted EBITDA 6.036M less D&A 9.789M plus 0.012M disposal gain reconciles to adjusted OP. Cash was 41.596M; net debt including leases was 12.343M.

September 7 interim financial statements and management outlook were reviewed against the May 28 FY2025 release. Full webcast Q&A transcripts were not obtained; this is a filings-and-management-release review. The latest outlook retains FY2026 revenue growth above 30% and low-teens adjusted EBITDA, not operating profit.

Latest interim and annual statements plus management releases were reviewed. Complete webcast Q&A transcripts were not obtained, and this is explicitly not claimed as two full earnings-call reads. Product-level revenue, costs, precise capacity and the complete historical option schedule remain evidence gaps.

## 4. Local evidence and calibration packet

H1 accounts replace a preliminary revenue-only anchor with disclosed GP, operating loss, D&A and cash. The model starts in H2, so both first rows are forecasts. H2 sales 34M/37M and OP-1.5M/-1M reconcile with an approximately 13.325M FY adjusted EBITDA case when H2 D&A equals H1. This meets the low-teens EBITDA direction without treating EBITDA as operating profit. The going-concern planning assumptions are distinguished from formal long-term public guidance.

## 5. Summing thesis score

| Category | Points / maximum | Evidence |
|---|---|---|
| Core Thesis | 14/20 | Customer financing and signed supply agreements validate capability, with competing epi suppliers. |
| Cycle | 8/10 | Recovery is printing, but adjusted operating losses remain. |
| TAM | 9/10 | InP, QDL, GaN power/RF and microLED expand applications without implying full end-market capture. |
| Catalysts | 4/5 | InP tooling conversion and customer sampling; prior sale catalyst has ended. |
| Macro | 9/10 | AI optics and defence demand support qualified epitaxy. |
| Alpha | 3/5 | Margin recovery opportunity, tempered by public awareness of the financing and supply deals. |
| Risks | -5 | Qualification/yield, customer concentration, substrate supply, margin/cash conversion and dilution. |

**Sum: 42/60 × 100 = 70.000000%, conventionally rounded to 70%.** This advisory thesis score does not replace any portfolio Base override. The separate risk classes below total the displayed deduction; holding status and classification do not waive that deduction.

## 6. Revenue sources, capture and product margins

Calendar quarters. Keep 20× adjusted operating profit and multiply GBP/share by 100 for GBp. H2 2026 revenue is GBP 34M/37M, giving FY2026 GBP 135.558M, about 39.3% growth. H2 OP is -1.5M/-1.0M, not breakeven. With H2 D&A assumed equal to H1 (9.789M), FY adjusted EBITDA is about 13.325M, within low teens. Product GP includes manufacturing depreciation; corporate operating costs include relevant administrative amortisation. No D&A add-back enters the valuation. The H2 InP/other-photonics/wireless allocations are analyst proxies, calibrated to those OP assumptions. Annual revenue vectors for 2027-30: InP 80/145/215/285M, other photonics 42/48/54/60M, wireless 58/64/71/78M. Weights 22/24/26/28%. Product GM ramps only as displayed, with InP easing from 38% in 2029 to 35% in 2030. Corporate/operating expense is GBP 22/26/30/34M annually. These are conditional high-utilization economics, not management guidance. Tower and MACOM sales are within InP; QDL, storage and other optical technologies are within other photonics, with no additional contract layer. The share proxy adds full MACOM conversion and June awards to September 30 non-treasury issued shares. Associated MACOM warrants expire on conversion: do not automatically count both as simultaneous dilution. The complete historical option inventory remains unresolved; the denominator is a specified minimum forward dilution scenario rather than a complete accounting diluted share count.

| Vector | Actual calibration / source | Forward demand and capture | Margin derivation / constraint |
|---|---|---|---|
| InP optical epitaxy | 12M Q3 allocation within 21M modeled photonics; product actual split is not disclosed. | Annual 80/145/215/285M GBP 2027-30. The 3.5625×2030/2027 bridge requires substantially more qualified output and richer content. At a purely illustrative 1,000 GBP per wafer-equivalent invoice,80k→285k units are required; actual wafer diameter, process scope and ASP remain unknown. | Quarterly 24/27/30/33% GM in 2027;36% 2028,38% 2029,35% 2030. Rising loading reduces fixed cost per sale, followed by modest price/mix normalization. At 1,000 GBP invoice and 36% GM,640 GBP productcost is the implied requirement, not disclosed unit cost. |
| Other photonics | 9M initial allocation; includes IR/VCSEL/storage and QDL. | 42/48/54/60M annual sales. The 14M USD multi-year storage order is included within this envelope, not counted each year or relabeled InP. QDL sampling supports a prospective mix shift but has no disclosed material revenue schedule. | 16% 2027→22% 2029-30 product GM proxy. New products may have lower initial yields; mature applications offset some ramp costs. |
| Wireless | 13M initial quarter; H1 reported 26.023M. | 58/64/71/78M annual sales;1.345×2027-30 recovery in qualified RF demand and content. No whole GaN power-market capture is assumed. | 6% 2027→10% 2029-30 product GM reflects recovery from low loading. This remains the lower-margin business in the model. |

**Confidence:** reported historical totals are HIGH. Source-based forward extrapolations are MEDIUM only where the cited source sizes the relevant period; other product allocations, future capture and margin assumptions are LOW. A source for the historical anchor does not make an unsourced future factor management guidance. The detailed factors below are the single modeled base case, with uncertain assumptions exposed for review.

## V1. InP optical epitaxy (within Photonics)

Sources: I1/2/3; InP customer agreements and assumed throughput/content. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

| Quarter | Revenue anchor | Demand / conversion factor | Revenue | Product margin | GP / contribution |
|---|---|---|---|---|---|
| Q3 2026 | 12.000000 | 1.000000 | 12.000 | 17.083% | 2.050 |
| Q4 2026 | 12.000000 | 1.166667 | 14.000 | 18.000% | 2.520 |
| Q1 2027 | 12.000000 | 1.466667 | 17.600 | 24.000% | 4.224 |
| Q2 2027 | 12.000000 | 1.600000 | 19.200 | 27.000% | 5.184 |

| Quarter | Revenue anchor | Demand / conversion factor | Revenue | Product margin | GP / contribution |
|---|---|---|---|---|---|
| Q3 2027 | 12.000000 | 1.733333 | 20.800 | 30.000% | 6.240 |
| Q4 2027 | 12.000000 | 1.866667 | 22.400 | 33.000% | 7.392 |
| Q1 2028 | 12.000000 | 2.658333 | 31.900 | 36.000% | 11.484 |
| Q2 2028 | 12.000000 | 2.900000 | 34.800 | 36.000% | 12.528 |
| Q3 2028 | 12.000000 | 3.141667 | 37.700 | 36.000% | 13.572 |
| Q4 2028 | 12.000000 | 3.383333 | 40.600 | 36.000% | 14.616 |
| Q1 2029 | 12.000000 | 3.941667 | 47.300 | 38.000% | 17.974 |
| Q2 2029 | 12.000000 | 4.300000 | 51.600 | 38.000% | 19.608 |
| Q3 2029 | 12.000000 | 4.658333 | 55.900 | 38.000% | 21.242 |
| Q4 2029 | 12.000000 | 5.016667 | 60.200 | 38.000% | 22.876 |
| Q1 2030 | 12.000000 | 5.225000 | 62.700 | 35.000% | 21.945 |
| Q2 2030 | 12.000000 | 5.700000 | 68.400 | 35.000% | 23.940 |
| Q3 2030 | 12.000000 | 6.175000 | 74.100 | 35.000% | 25.935 |
| Q4 2030 | 12.000000 | 6.650000 | 79.800 | 35.000% | 27.930 |

## V2. Other Photonics: IR, VCSEL, QDL and storage

Sources: I1/4/5; IR/VCSEL base plus storage and QDL within the same allocation. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

| Quarter | Revenue anchor | Demand / conversion factor | Revenue | Product margin | GP / contribution |
|---|---|---|---|---|---|
| Q3 2026 | 9.000000 | 1.000000 | 9.000 | 12.000% | 1.080 |
| Q4 2026 | 9.000000 | 1.055556 | 9.500 | 12.000% | 1.140 |
| Q1 2027 | 9.000000 | 1.026667 | 9.240 | 16.000% | 1.478 |
| Q2 2027 | 9.000000 | 1.120000 | 10.080 | 16.000% | 1.613 |
| Q3 2027 | 9.000000 | 1.213333 | 10.920 | 16.000% | 1.747 |

| Quarter | Revenue anchor | Demand / conversion factor | Revenue | Product margin | GP / contribution |
|---|---|---|---|---|---|
| Q4 2027 | 9.000000 | 1.306667 | 11.760 | 16.000% | 1.882 |
| Q1 2028 | 9.000000 | 1.173333 | 10.560 | 20.000% | 2.112 |
| Q2 2028 | 9.000000 | 1.280000 | 11.520 | 20.000% | 2.304 |
| Q3 2028 | 9.000000 | 1.386667 | 12.480 | 20.000% | 2.496 |
| Q4 2028 | 9.000000 | 1.493333 | 13.440 | 20.000% | 2.688 |
| Q1 2029 | 9.000000 | 1.320000 | 11.880 | 22.000% | 2.614 |
| Q2 2029 | 9.000000 | 1.440000 | 12.960 | 22.000% | 2.851 |
| Q3 2029 | 9.000000 | 1.560000 | 14.040 | 22.000% | 3.089 |
| Q4 2029 | 9.000000 | 1.680000 | 15.120 | 22.000% | 3.326 |
| Q1 2030 | 9.000000 | 1.466667 | 13.200 | 22.000% | 2.904 |
| Q2 2030 | 9.000000 | 1.600000 | 14.400 | 22.000% | 3.168 |
| Q3 2030 | 9.000000 | 1.733333 | 15.600 | 22.000% | 3.432 |
| Q4 2030 | 9.000000 | 1.866667 | 16.800 | 22.000% | 3.696 |

## V3. Wireless epitaxy

Sources: I1/7; wireless recovery and qualified product demand. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

| Quarter | Revenue anchor | Demand / conversion factor | Revenue | Product margin | GP / contribution |
|---|---|---|---|---|---|
| Q3 2026 | 13.000000 | 1.000000 | 13.000 | 4.000% | 0.520 |
| Q4 2026 | 13.000000 | 1.038462 | 13.500 | 4.000% | 0.540 |
| Q1 2027 | 13.000000 | 0.981538 | 12.760 | 6.000% | 0.766 |
| Q2 2027 | 13.000000 | 1.070769 | 13.920 | 6.000% | 0.835 |
| Q3 2027 | 13.000000 | 1.160000 | 15.080 | 6.000% | 0.905 |
| Q4 2027 | 13.000000 | 1.249231 | 16.240 | 6.000% | 0.974 |

| Quarter | Revenue anchor | Demand / conversion factor | Revenue | Product margin | GP / contribution |
|---|---|---|---|---|---|
| Q1 2028 | 13.000000 | 1.083077 | 14.080 | 8.000% | 1.126 |
| Q2 2028 | 13.000000 | 1.181538 | 15.360 | 8.000% | 1.229 |
| Q3 2028 | 13.000000 | 1.280000 | 16.640 | 8.000% | 1.331 |
| Q4 2028 | 13.000000 | 1.378462 | 17.920 | 8.000% | 1.434 |
| Q1 2029 | 13.000000 | 1.201538 | 15.620 | 10.000% | 1.562 |
| Q2 2029 | 13.000000 | 1.310769 | 17.040 | 10.000% | 1.704 |
| Q3 2029 | 13.000000 | 1.420000 | 18.460 | 10.000% | 1.846 |
| Q4 2029 | 13.000000 | 1.529231 | 19.880 | 10.000% | 1.988 |
| Q1 2030 | 13.000000 | 1.320000 | 17.160 | 10.000% | 1.716 |
| Q2 2030 | 13.000000 | 1.440000 | 18.720 | 10.000% | 1.872 |
| Q3 2030 | 13.000000 | 1.560000 | 20.280 | 10.000% | 2.028 |
| Q4 2030 | 13.000000 | 1.680000 | 21.840 | 10.000% | 2.184 |

## 7. Operating-cost bridge and quarterly operating model

Explicit annual expenses 22/26/30/34M GBP 2027-30 rise despite a declining ratio. Product margins include manufacturing depreciation and relevant amortization remains in operating costs. H1 adjusted EBITDA6.036M-D&A9.789M+disposal 0.012M=-3.741M adjusted OP. Adjusted OP itself excludes certain charges that may recur; it is not free cash flow. Future reactor conversion and customer qualification require investment in addition to the accounting profit bridge.

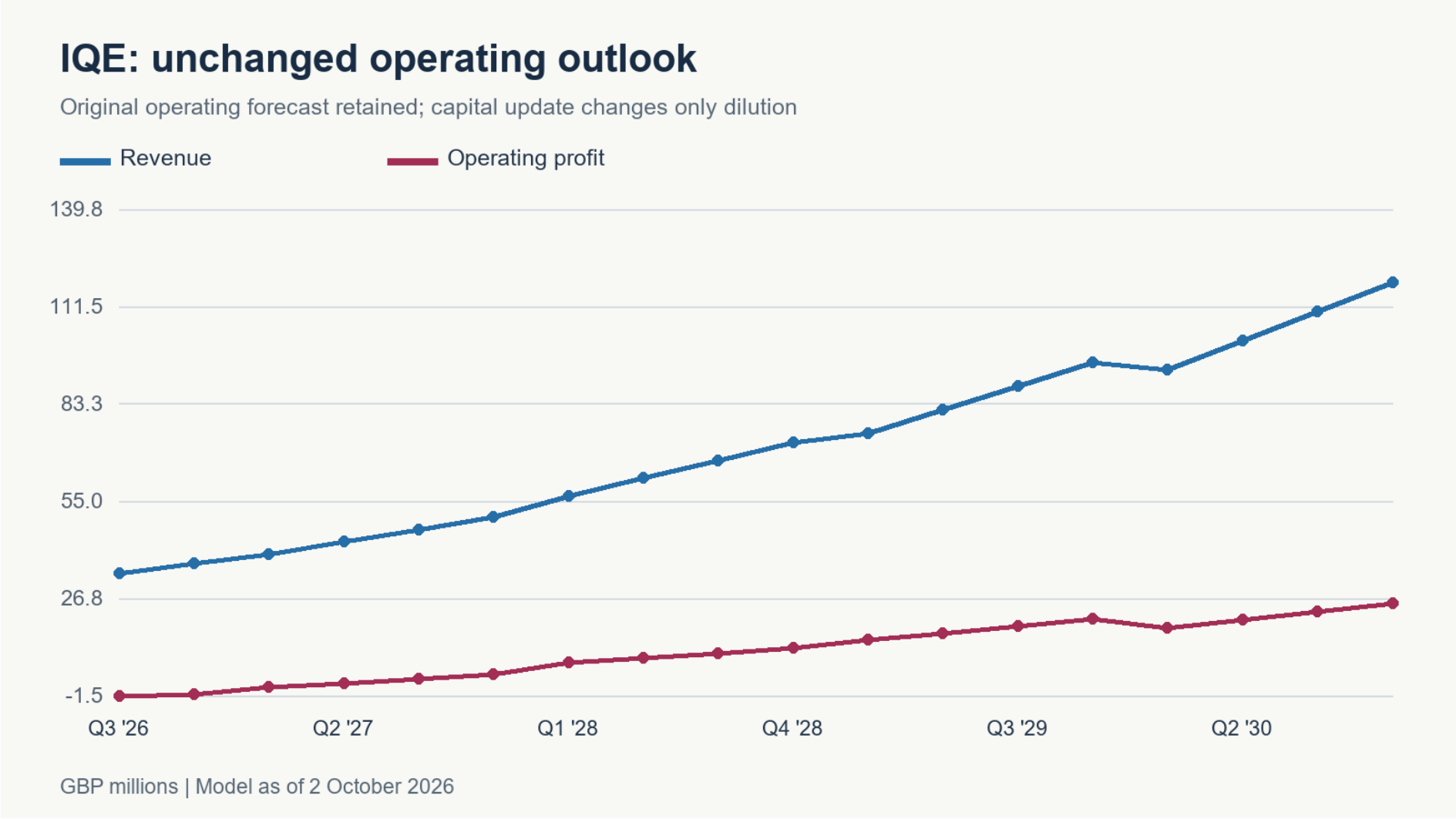
**Profit basis:** Adjusted operating profit after depreciation and amortisation. Amounts below are GBP millions.

| Quarter | Status | Revenue | GP / contribution | Opex % | Opex | OP margin | Operating profit | Main driver | Confidence |
|---|---|---|---|---|---|---|---|---|---|
| Q3 2026 | Model | 34.000 | 3.650 | 15.147% | 5.150 | -4.412% | -1.500 (neg) | H2 InP ramp with operating loss | LOW model |
| Q4 2026 | Model | 37.000 | 4.200 | 14.054% | 5.200 | -2.703% | -1.000 (neg) | Higher sales; D&A still material | LOW model |

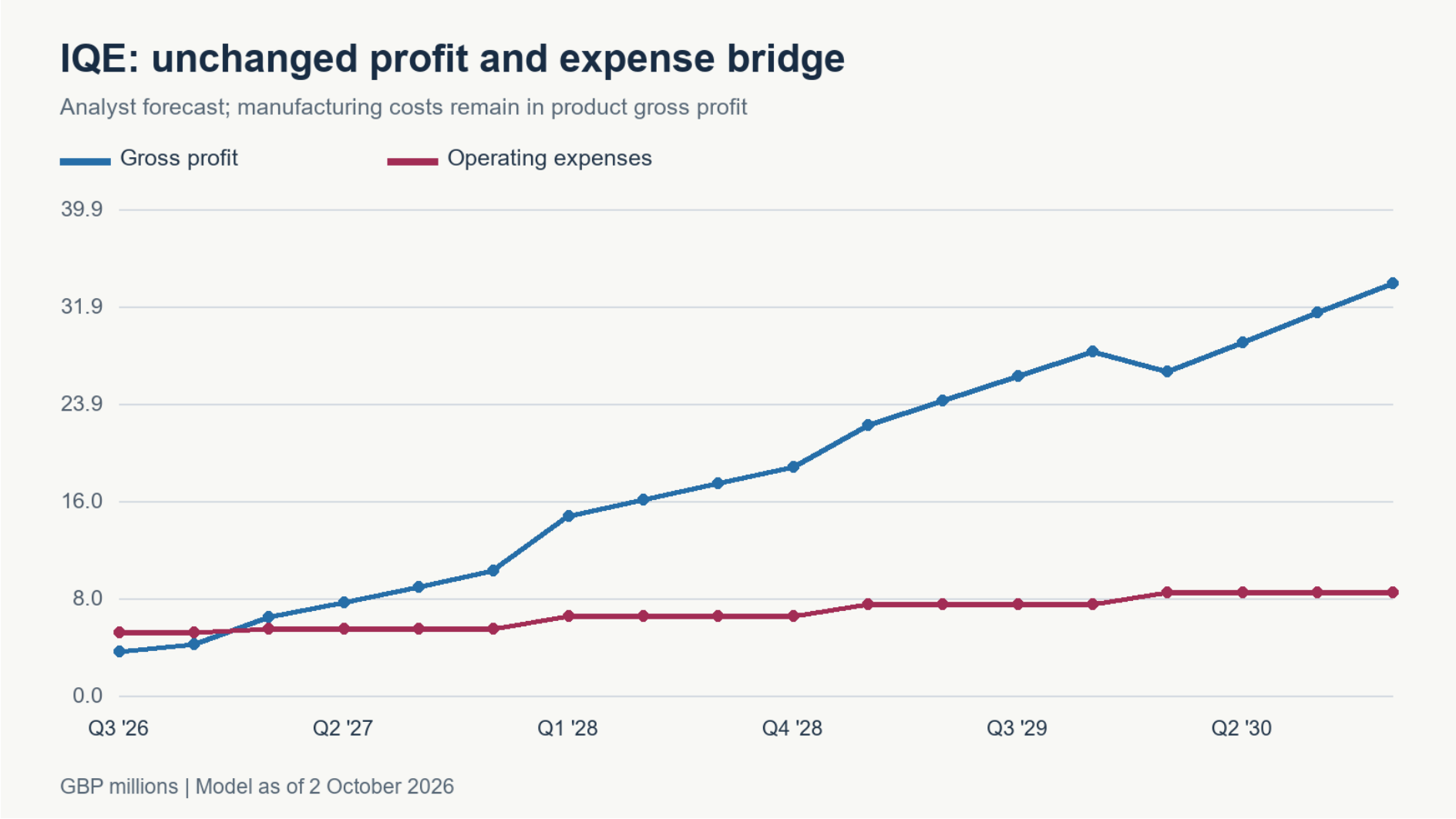
| Quarter | Status | Revenue | GP / contribution | Opex % | Opex | OP margin | Operating profit | Main driver | Confidence |
|---|---|---|---|---|---|---|---|---|---|
| Q1 2027 | Model | 39.600 | 6.468 | 13.889% | 5.500 | 2.444% | 0.968 | 2027 qualified volume | LOW model |
| Q2 2027 | Model | 43.200 | 7.632 | 12.731% | 5.500 | 4.935% | 2.132 | InP loading improves | LOW model |
| Q3 2027 | Model | 46.800 | 8.892 | 11.752% | 5.500 | 7.248% | 3.392 | Higher InP product margin | LOW model |
| Q4 2027 | Model | 50.400 | 10.248 | 10.913% | 5.500 | 9.421% | 4.748 | 2027 exit production | LOW model |
| Q1 2028 | Model | 56.540 | 14.722 | 11.496% | 6.500 | 14.543% | 8.222 | 2028 capacity conversion | LOW model |
| Q2 2028 | Model | 61.680 | 16.061 | 10.538% | 6.500 | 15.501% | 9.561 | More qualified epi output | LOW model |
| Q3 2028 | Model | 66.820 | 17.399 | 9.728% | 6.500 | 16.311% | 10.899 | Customer adoption | LOW model |
| Q4 2028 | Model | 71.960 | 18.738 | 9.033% | 6.500 | 17.006% | 12.238 | 2028 mix and throughput | LOW model |
| Q1 2029 | Model | 74.800 | 22.150 | 10.027% | 7.500 | 19.585% | 14.650 | 2029 higher optical content | LOW model |
| Q2 2029 | Model | 81.600 | 24.163 | 9.191% | 7.500 | 20.421% | 16.663 | Strong utilization case | LOW model |
| Q3 2029 | Model | 88.400 | 26.177 | 8.484% | 7.500 | 21.128% | 18.677 | Peak modeled product margin | LOW model |
| Q4 2029 | Model | 95.200 | 28.190 | 7.878% | 7.500 | 21.734% | 20.690 | 2029 production scale | LOW model |
| Q1 2030 | Model | 93.060 | 26.565 | 9.134% | 8.500 | 19.412% | 18.065 | 2030 price/mix normalization | LOW model |
| Q2 2030 | Model | 101.520 | 28.980 | 8.373% | 8.500 | 20.173% | 20.480 | Continued volume growth | LOW model |
| Q3 2030 | Model | 109.980 | 31.395 | 7.729% | 8.500 | 20.817% | 22.895 | Broader photonics demand | LOW model |

| Quarter | Status | Revenue | GP / contribution | Opex % | Opex | OP margin | Operating profit | Main driver | Confidence |
|---|---|---|---|---|---|---|---|---|---|
| Q4 2030 | Model | 118.440 | 33.810 | 7.177% | 8.500 | 21.369% | 25.310 | 2030 steady conversion case | LOW model |

OP = sum of product GP/contribution - explicit operating costs. OP margin = OP ÷ revenue. Displayed numbers are rounded for reading; the workbook calculates from unrounded inputs. Guidance assigned to a future quarter retains Model status because guidance is not an actual result.




*IQE: unchanged operating outlook. GBP millions. Original operating forecast retained; capital update changes only dilution*




*IQE: unchanged profit and expense bridge. GBP millions. Analyst forecast; manufacturing costs remain in product gross profit*

## 8. Multiple, shares and recognition convention

20×forward adjusted OP reflects qualified compound-semiconductor recipes, optical demand and customer commitments. It is conditional on operating conversion after depreciation. Competitors and capital intensity prevent importing finished-device or memory-vendor margins wholesale. There is no takeover premium or implied strategic-investor price floor.

September 30 issued 1,331.131442M minus 0.004115M treasury, plus GBP15M / GBP0.198 = 75.757575758M MACOM conversion shares, plus unchanged 5.300788M June employee awards gives 1,412.185690758M. Actual issued shares rose 0.036293M from August; only that increment changes the existing forward proxy. It is not a complete accounting option inventory. Associated MACOM warrants expire on conversion and are not counted again. Cash41.596M and net debt including leases12.343M remain the historical H1 anchors. [I1/I2/I8] Future awards can exceed this floor, which the separate share sensitivity exposes.

The valuation denominator is the ceiling quarter's diluted-share proxy. It is not a historical EPS weighted-average assumption disguised as a fully known future capitalization. Unit factor: 100; this converts the model units to GBp per share. No cash balance, enterprise-value adjustment or after-tax earnings substitution is added to the operating-profit ceiling.

## Actual capital update and unchanged forward allowances

The September 30 RNS reports 1,331,131,442 ordinary shares of1p each, including4,115 treasury shares, leaving1,331,127,327 voting rights. The reported issue count is not a weighted-average diluted EPS denominator. The RNS does not specify which instrument caused the36,293-share increase; this update changes only actual issued capital and retains every existing forward allowance. The unchanged June-award allowance is not recast as newly issued equity. [I8]

| Input / output | Prior model | Updated model |
|---|---|---|
| Issued ordinary shares, M | 1,331.095149 | 1,331.131442 |
| Treasury shares, M | 0.004115 | 0.004115 |
| MACOM full conversion, M | 75.757575758 | 75.757575758 |
| June award allowance, M | 5.300788 | 5.300788 |
| Forward share proxy, M | 1,412.149397758 | 1,412.185690758 |
| Q4 2027 F4Q OP, GBP M | 33.430400 | 33.430400 |
| Q4 2027 ceiling, GBp | 47.346831791 | 47.345614984 |
| Capital-only change | +0.036293M shares | -0.001216808 GBp / -0.002570% |

## 9. Complete rolling four-quarter ceiling derivation




*IQE: rolling four-quarter price-target path. GBp per share. Lines nearly overlap; exact capital-only changes are tabulated below*

| Ceiling quarter | Exact earnings window | Forward 4Q OP | Multiple | Diluted shares M | Ceiling GBp | Key inflection |
|---|---|---|---|---|---|---|
| Q3 2026 | Q3 2026 to Q2 2027 | 0.600000 | 20x | 1,412.185691 | 0.849747 | H2 InP ramp with operating loss |
| Q4 2026 | Q4 2026 to Q3 2027 | 5.492000 | 20x | 1,412.185691 | 7.778014 | Higher sales; D&A still material |

| Ceiling quarter | Exact earnings window | Forward 4Q OP | Multiple | Diluted shares M | Ceiling GBp | Key inflection |
|---|---|---|---|---|---|---|
| Q1 2027 | Q1 2027 to Q4 2027 | 11.240000 | 20x | 1,412.185691 | 15.918586 | 2027 qualified volume |
| Q2 2027 | Q2 2027 to Q1 2028 | 18.494400 | 20x | 1,412.185691 | 26.192589 | InP loading improves |
| Q3 2027 | Q3 2027 to Q2 2028 | 25.923200 | 20x | 1,412.185691 | 36.713585 | Higher InP product margin |
| Q4 2027 | Q4 2027 to Q3 2028 | 33.430400 | 20x | 1,412.185691 | 47.345615 | 2027 exit production |
| Q1 2028 | Q1 2028 to Q4 2028 | 40.920000 | 20x | 1,412.185691 | 57.952719 | 2028 capacity conversion |
| Q2 2028 | Q2 2028 to Q1 2029 | 47.347200 | 20x | 1,412.185691 | 67.055204 | More qualified epi output |
| Q3 2028 | Q3 2028 to Q2 2029 | 54.449600 | 20x | 1,412.185691 | 77.113938 | Customer adoption |
| Q4 2028 | Q4 2028 to Q3 2029 | 62.227200 | 20x | 1,412.185691 | 88.128920 | 2028 mix and throughput |
| Q1 2029 | Q1 2029 to Q4 2029 | 70.680000 | 20x | 1,412.185691 | 100.100150 | 2029 higher optical content |
| Q2 2029 | Q2 2029 to Q1 2030 | 74.095400 | 20x | 1,412.185691 | 104.937191 | Strong utilization case |
| Q3 2029 | Q3 2029 to Q2 2030 | 77.912200 | 20x | 1,412.185691 | 110.342713 | Peak modeled product margin |
| Q4 2029 | Q4 2029 to Q3 2030 | 82.130400 | 20x | 1,412.185691 | 116.316715 | 2029 production scale |
| Q1 2030 | Q1 2030 to Q4 2030 | 86.750000 | 20x | 1,412.185691 | 122.859197 | 2030 price/mix normalization |

- **Q3 2026:** (-1.500000 + -1.000000 + 0.968000 + 2.132000) x20 x100 / 1,412.185690758 = **0.849747 GBp/share**. Window: Q3 2026 to Q2 2027.
- **Q4 2026:** (-1.000000 + 0.968000 + 2.132000 + 3.392000) x20 x100 / 1,412.185690758 = **7.778014 GBp/share**. Window: Q4 2026 to Q3 2027.
- **Q1 2027:** (0.968000 + 2.132000 + 3.392000 + 4.748000) x20 x100 / 1,412.185690758 = **15.918586 GBp/share**. Window: Q1 2027 to Q4 2027.
- **Q2 2027:** (2.132000 + 3.392000 + 4.748000 + 8.222400) x20 x100 / 1,412.185690758 = **26.192589 GBp/share**. Window: Q2 2027 to Q1 2028.
- **Q3 2027:** (3.392000 + 4.748000 + 8.222400 + 9.560800) x20 x100 / 1,412.185690758 = **36.713585 GBp/share**. Window: Q3 2027 to Q2 2028.
- **Q4 2027:** (4.748000 + 8.222400 + 9.560800 + 10.899200) x20 x100 / 1,412.185690758 = **47.345615 GBp/share**. Window: Q4 2027 to Q3 2028.
- **Q1 2028:** (8.222400 + 9.560800 + 10.899200 + 12.237600) x20 x100 / 1,412.185690758 = **57.952719 GBp/share**. Window: Q1 2028 to Q4 2028.
- **Q2 2028:** (9.560800 + 10.899200 + 12.237600 + 14.649600) x20 x100 / 1,412.185690758 = **67.055204 GBp/share**. Window: Q2 2028 to Q1 2029.
- **Q3 2028:** (10.899200 + 12.237600 + 14.649600 + 16.663200) x20 x100 / 1,412.185690758 = **77.113938 GBp/share**. Window: Q3 2028 to Q2 2029.
- **Q4 2028:** (12.237600 + 14.649600 + 16.663200 + 18.676800) x20 x100 / 1,412.185690758 = **88.128920 GBp/share**. Window: Q4 2028 to Q3 2029.
- **Q1 2029:** (14.649600 + 16.663200 + 18.676800 + 20.690400) x20 x100 / 1,412.185690758 = **100.100150 GBp/share**. Window: Q1 2029 to Q4 2029.
- **Q2 2029:** (16.663200 + 18.676800 + 20.690400 + 18.065000) x20 x100 / 1,412.185690758 = **104.937191 GBp/share**. Window: Q2 2029 to Q1 2030.
- **Q3 2029:** (18.676800 + 20.690400 + 18.065000 + 20.480000) x20 x100 / 1,412.185690758 = **110.342713 GBp/share**. Window: Q3 2029 to Q2 2030.

- **Q4 2029:** (20.690400 + 18.065000 + 20.480000 + 22.895000) x20 x100 / 1,412.185690758 = **116.316715 GBp/share**. Window: Q4 2029 to Q3 2030.
- **Q1 2030:** (18.065000 + 20.480000 + 22.895000 + 25.310000) x20 x100 / 1,412.185690758 = **122.859197 GBp/share**. Window: Q1 2030 to Q4 2030.

The final three operating quarters have no ceiling because a complete four-quarter window is unavailable. No OP, cost, multiple, unit-factor or probability input changes in this capital update.

## 10. Annual trajectory cross-check

| Calendar year | Revenue | GP / contribution | Opex | OP | OP margin |
|---|---|---|---|---|---|
| 2027 | 180.000 | 33.240 | 22.000 | 11.240 | 6.244% |
| 2028 | 257.000 | 66.920 | 26.000 | 40.920 | 15.922% |
| 2029 | 340.000 | 100.680 | 30.000 | 70.680 | 20.788% |
| 2030 | 423.000 | 120.750 | 34.000 | 86.750 | 20.508% |

These are calendar-year sums of the quarterly path, not extra revenue and not the live valuation method. For non-December fiscal reporters, fiscal guidance and these calendar totals must be aligned using the period mapping above; a four-quarter run rate is not automatically the same fiscal year. Outer-year outputs are conditional TAM/capture trajectories, not verified management sales targets.

Annual company sales 180M/257M/340M/423M GBP 2027-30.2027 growth is 32.8% over modeled 2026 sales 135.558M, consistent with rather than guaranteed by the internal planning assumptions. InP becomes 67.4% of 2030 sales, so the case depends on mix change and productive capacity, not uniform growth across all businesses. Customer agreements lack enough public quantities to independently validate 285M InP sales.

## 11. Dilution sensitivity - same operating case

| Denominator sensitivity | Q4 2027 shares M | Q4 2027 ceiling GBp | Effect on ceiling |
|---|---|---|---|
| -5% shares | 1,341.576406 | 49.837489 | +5.26% |
| +0% shares | 1,412.185691 | 47.345615 | +0.00% |
| +5% shares | 1,482.794975 | 45.091062 | -4.76% |
| +10% shares | 1,553.404260 | 43.041468 | -9.09% |

This isolates denominator uncertainty; it is not a second operating case or permission to remove known dilution. The updated share path above remains the active convention. The comparison now uses the current Q4 2027 horizon; operating assumptions are unchanged.

## 12. Programme and probability ledger

| Programme / signal | Source and probability | Amount / unit base | Treatment in this model | Escalation / recognition |
|---|---|---|---|---|
| Tower supply | Signed customer agreement;85% programme confidence | Minimum commitments with undisclosed amounts | Inside InP; no additive contract windfall | Paid qualified wafer shipments and disclosed quantities calibrate capture. |
| MACOM capacity | Customer financing;85% deployment confidence | Known funding, unknown detailed revenue schedule | Inside InP and share bridge | Installed/qualified tooling and net sales determine conversion. |
| Storage order | Announced multi-year production order;85% execution confidence | 14M USD total over multiple years | Inside other photonics, no annual repetition | Delivery and recognition schedule replaces the allocation. |
| QDL sampling | Named customer purchase agreement;60% material-scale confidence | No public mass-production value | Supports other-photonics capture; zero separate optionality amount | Customer launch and recurring orders upgrade the financial case. |
| Outer InP ramp | Analyst demand/capture scenario;50% confidence | 3×wafer-equivalent volume×1.1875 content | Embedded long-range base path, not a second haircut | Disclosed reactor capacity, yield and ASP must support output. |

Probabilities are analyst judgments under the framework signal hierarchy. Embedded revenue is not multiplied or added a second time. Where no defensible amount or unit base exists, the opportunity remains explicitly unquantified; confidence alone cannot create a revenue number. New evidence can enlarge the TAM, improve capture or move recognition earlier, but each requires an identifiable model input.

## 13. Scored risk classes

| Risk class | Scored deduction | Mechanism and consequence |
|---|---|---|
| Qualification and yield | -1 | A reactor or sample is not qualified sellable production; utilization losses consume cash. |
| Customer concentration | -1 | Funding and purchase commitments can increase customer bargaining power. |
| Substrate supply | -1 | InP input availability, quality and restrictions can limit epi output despite demand. |
| Margin and cash conversion | -1 | Manufacturing depreciation, leases and recurring adjustments limit economic profit. |
| Dilution | -1 | Convertible shares and employee awards reduce common claims; the option inventory is incomplete. |

## 14. Dated catalysts, counters and escalation

1. **2026-12-31 - FY26 production** Test H2 sales 71M and continuing operating losses against actual shipments and D&A.
2. **2027-03-31 monitoring date - Annual results** Reconcile low-teens EBITDA to OP and cash; refresh all 18 operating quarters.
3. **2027-06-30 - InP conversion** Verify qualified output and actual photonics mix, not installed reactors alone.
4. **2027-12-31 - Customer scale** Check 180M company sales and rising product GP against spending.
5. **2028-12-31 - High-utilization economics** 36% InP GM requires actual cost absorption, not a shortage narrative alone.
6. **Every capital update - Conversion and awards** Rebuild complete dilution if new instruments, awards or conversion-price changes appear.

## 15. Cross-book monitoring

The Tower agreement is a verified supplier/customer link. IQE epitaxy is an input to the downstream foundry platform, so the two companies' revenue does not represent independent end-customer capex. InP and GaAs QDL are different technology routes; no assumption that each captures all optical lanes is made. CRDO/MRVL optical adoption expands potential device demand; IQE's wafer revenue still requires a specific qualified process/customer route. Tower's larger SiPho outlook cannot be multiplied by an arbitrary IQE take rate without specifying the InP content, architecture and sourcing share.

## 16. Source register and model limits

**I1 - H1 2026 results and notes (2026-09-07).** Notes 7-8 reconcile adjusted GP, EBITDA and OP. The going-concern model assumes at least 30% revenue growth in 2026 and 2027; that internal planning assumption is not a separate long-term public earnings target.

**I2 - Strategic review conclusion (2026-04-27).** The review concluded with financing. The prior artifact's pending takeover/Taiwan-sale catalyst is removed. Customer investment is not a contractual floor for public shares.

**I3 - Tower supply agreement (2026-06-15).** Minimum purchases/volumes and reciprocal supply commitments were announced, without disclosed revenue amounts. The agreement also settles the prior IP dispute.

**I4 - Production order (2026-07-14).** The $14M order is multi-year and supports storage applications. It is not $14M annual InP revenue.

**I5 - Quintessent QDL agreement (2026-09-03).** Six-inch GaAs QDL epitaxy is progressing to customer sampling; the release does not quantify a production revenue ramp.

**I6 - August voting rights (2026-09-01).** Historical August baseline: issued1,331.095149M less0.004115M treasury. Replaced in the active bridge by September30 count I8; MACOM and award allowances retained.

**I7 - FY2025 release (2026-05-28).** FY2025 revenue GBP 97.3M anchors the FY2026 growth comparison.

**I8 - September total voting rights, RNS0275X (2026-09-30).** Full original story linked through the issuer regulatory-news page:1,331,131,442 ordinary1p shares,4,115 treasury and1,331,127,327 total voting rights as at September30. Actual issued increment36,293 from August. No new financing or employee-award forecast is inferred from that count.

No complete two-webcast Q&A review. Product revenue splits, margins, quarterly timing and outer-year expansion are LOW-confidence assumptions. The disclosed multi-year agreements lack a complete revenue schedule. Full future option dilution is not reconstructed. Non-GAAP OP is not cash flow and the framework multiple does not separately adjust for leases or cash.

**Workbook:** v3_2_master_portfolio.xlsx, model tab IQE_L_Q_Model. It owns portfolio ranking, prices and return calculations; those are intentionally outside this artifact. **Current artifact path:** G:\My Drive\1amInvesting\Artifacts\IQEL_IQE_DeepDive.md. **Original snapshot:** versions/originals/IQEL_IQE_DeepDive.md. Superseded history is retained separately in VERSION_LOG, not in this current-state model.